

Edge re-authoring pass — final report

AI Supply Chain Terminal

2026-07-21

1. What this pass did

Re-authored 53 edges — the 47 previously rescaled + the gallium/RF override + 5 adjacent HBM input_to edges that carried the same rescaling problem — from the paper. Not from arithmetic. Post-pass, no `supplies` or `input_to` bucket sums to exactly 1.0 by construction. Bucket incompleteness is the deliverable, not a bug.

Populated one `output_share` (HBM → NVIDIA = 0.90) from the paper's own figure on the paper's own edge.

Assertions:

```
OK A1 no supplies / input_to bucket sums to exactly 1.0 by construction
OK A2 every changed input_share has a source_note
OK A3 every output_share has a source_note (1 / 1 populated)
OK A4 edge count unchanged (167)
OK A5 63 nodes score under both normalize modes
OK A6 no inbound_hhi > 1.0 under either mode (norm: 0, unnorm: 0)
```

2. Re-authoring ledger — 53 edges

Every value has a paper-cited or explicit “no direct paper basis; industry-informed estimate” source note on the edge. Grouped by target-bucket:

company:tsmc supplies

Paper §2D anchors ASML as sole EUV maker. Paper does not quantify per-supplier share of a foundry's equipment input; the values below are industry-informed estimates of leading-edge fab tool-spend mix.

edge	old	new
ASML → TSMC	0.514	0.400
AMAT → TSMC	0.171	0.100
LRCX → TSMC	0.114	0.080
KLA → TSMC	0.086	0.050
TEL → TSMC	0.114	0.080
bucket sum	1.000	0.710

company:samsung supplies

Same rationale; slightly different mix because Samsung's memory business raises AMAT / LRCX share.

edge	old	new
ASML → Samsung	0.250	0.200
AMAT → Samsung	0.250	0.120
LRCX → Samsung	0.208	0.120
KLA → Samsung	0.125	0.060
TEL → Samsung	0.167	0.100
bucket sum	1.000	0.600

product:hbm input_to (per-buyer BOM share)

Paper §2C names HBM a co-equal bottleneck with the GPU. Does not quantify HBM's cost share; industry-informed: HBM ~25–35% of AI-GPU BOM.

edge	old	new
HBM → NVIDIA	0.486	0.300
HBM → AMD	0.469	0.250
HBM → Broadcom	0.276	0.150
HBM → Google	0.350	0.150
HBM → Amazon	0.250	0.120
HBM → Microsoft	0.250	0.120
HBM → Meta	0.250	0.120

product:cowos_packaging input_to

Paper §2E: CoWoS repeatedly gated GPU output. Paper does not quantify cost share; industry: advanced packaging ~15–25% of AI-GPU cost.

edge	old	new
CoWoS → NVIDIA	0.513	0.200
CoWoS → AMD	0.531	0.180
CoWoS → Broadcom	0.621	0.150
CoWoS → Google	0.600	0.150

product:rf_power_semis input_to

edge	old	new
RF → Broadcom	0.103	0.100

mineral:gallium → product:rf_power_semis (paper override, restated)

Paper basis, restated independently of tier effect: Paper §1A: “*Gallium (GaAs/GaN wafers, RF chips, power electronics).*” RF & Power Semiconductors ARE gallium-material devices; the paper explicitly names gallium as the semiconductor input for this device class. Value stands on that paper claim alone.

edge	old	new
gallium → rf_power_semis	0.90	0.90

Hyperscaler supplies (chip supply into cloud/AI-lab operators)

Paper §2G names the hyperscaler / custom-silicon design pairings but does not quantify per-supplier share. Values are industry-informed estimates. Buckets deliberately incomplete.

edge	old	new
TSMC → Amazon	0.586	0.300
Marvell → Amazon	0.207	0.100
NVIDIA → Amazon	0.207	0.200
TSMC → Google	0.529	0.300
Broadcom → Google	0.294	0.200
NVIDIA → Google	0.176	0.100
TSMC → Meta	0.486	0.300
Broadcom → Meta	0.229	0.150
NVIDIA → Meta	0.286	0.250
TSMC → Microsoft	0.567	0.300
NVIDIA → Microsoft	0.333	0.300
Constellation → Microsoft	0.100	0.100

Data-centre facility supplies

Paper §3–4 name Vertiv (cooling leader), Siemens Energy / GE Vernova (grid equipment with multi-year backlogs), Constellation / NextEra / Vistra / Talen (utilities). Paper does not quantify per-facility share. Industry-informed estimates.

12 edges into facility:colossus / facility:stargate_abilene / facility:vantage_frontier at values 0.05–0.20 each. Full list in the edges.json diff.

3. output_share population

Only one edge populated, per spec ("populate only what the paper quantifies").

edge	value	paper basis
HBM → NVIDIA	0.90	Paper §5: "SK Hynix ~90% reliant on NVIDIA for HBM." Applied to HBM → NVIDIA as the closest existing edge in the two-hop path SK Hynix → HBM → NVIDIA. Previous 0.70 (world-HBM-share arithmetic across three makers, a constructed number) overwritten with the paper's own figure on the paper's own edge.

The previous pass's constructed 0.70 (world-HBM-share arithmetic across three makers) has been overwritten by the paper's stated 0.90.

4. Per-target sum table

type	targets	sum > 1.0	min	median	max
mines	5	1	0.65	1.00	1.17
refines	5	2	0.90	1.00	1.10
supplies	23	0	0.15	0.60	1.00
input_to	21	0	0.08	0.15	1.00
operates	5	0	1.00	1.00	1.00
located_in	8	2	1.00	1.00	6.00

Every `supplies` and `input_to` bucket sums to less than or equal to 1.0. Most are meaningfully below 1.0 — that is the honest reading, not a regression. The share-completeness backlog now correctly counts these as incomplete.

`mines` / `refines` / `located_in` sums over 1.0 come from country + facility double-counting that this task is scope-out to fix.

5. Full 63-node before/after — 1 tier change

Only one node moved on the default `normalize=true`.

node	old sev → new sev	old tier → new tier
ASML	0.244 → 0.202	critical → high

Every other tier is stable across this pass on the default measure. Neodymium's severity moved slightly (0.220 → same), a handful of low-severity nodes moved a few thousandths — all within the same tier.

The single tier change matters, so §6 investigates it.

6. ASML investigation — the required §4 explanation

Which axis moved: outbound. ASML has no inbound supply edges in the graph, so `inbound_hhi` = 0 under both modes. All of its severity comes from `outbound_criticality`.

Which specific edge changes: ASML supplies TSMC, Samsung, Intel, SK Hynix, Micron. Two of its five outbound edges were re-authored in this pass, both proportionally reduced from the previous rescaled values:

edge	pre-rescale (initial)	after rescale	after re-authoring
ASML → TSMC	0.900	0.514	0.400
ASML → Samsung	0.300	0.250	0.200
ASML → Intel	0.250	0.250	0.250
ASML → SK Hynix	0.150	0.150	0.150
ASML → Micron	0.100	0.100	0.100

Cascade + `outbound_criticality` still read `input_share` (per spec), so as ASML's `input_shares` on its most consequential downstream edge (TSMC) declined across two passes (0.900 → 0.514 → 0.400), ASML's `outbound_criticality` declined too.

pass	outbound_criticality	severity	tier
before per-stage HHI	0.732	0.393	critical
after edge-weight split	0.453	0.244	critical
after re-authoring	0.375	0.202	high

Position: sitting on the boundary. ASML is at severity 0.202, threshold is 0.225. It is 0.023 below critical — the smallest possible gap that produces a tier change. Not stable.

Deeper story — why this is happening.

The `input_share` on ASML → TSMC (0.40) represents ASML's share of TSMC's *total equipment input*. That is the correct reading under the two-field schema. But the paper's chokepoint story for ASML is not "ASML is 40% of TSMC's total tool spend" — it is "ASML is the sole EUV supplier, and EUV is required for leading-edge chips."

The concentration measure rewards *large fraction of a target's total supply*. It does not represent *sole supplier of a critical category*. Under the current data model, categories inside `supplies` (litho vs. deposition vs. etch vs. inspection) are not separated — every equipment supplier feeds one bucket. So ASML at 0.40 cannot outscore the combination of AMAT (0.10) + LRCX (0.08) + KLA (0.05) + TEL (0.08) on inbound HHI, and its outbound criticality reads the 0.40 edge, not the "if this one supplier disappears, TSMC's leading-edge output stops" story.

Three ways the model could represent this properly; none of them are in scope:

- Split `supplies` into per-category buckets (litho, deposition, etch, inspection). ASML's litho bucket would be a single-edge bucket at 1.0 and would win the `max`. Schema change.
- Have cascade read `output_share` (or a hybrid) so ASML's near-100% of TSMC's litho input is what propagates downstream. Behavioural change flagged as out of scope in the two-field split

spec.

- Introduce an explicit "sole supplier" / substitutability field on the edge itself. Schema change.

Not adjusting. The tier reading is honest given the current data model. ASML at high, 0.023 from critical, is the finding.

7. HBM status

Its critical → high drop persists.

pass	inbound	outbound	severity	tier
before edge split	0.440	0.566	0.234	critical
after edge split	0.440	0.424	0.178	high
after re-authoring	0.440	0.252	0.178	high

The concentration is `max(inbound, outbound) = 0.440` and has been all along — HBM's inbound is inbound-dominated. Its severity comes from three-supplier concentration (SK Hynix 0.60 + Micron 0.21 + Samsung 0.19, HHI = 0.44).

Restoring the paper's dependency judgment to `output_share` (0.90 on HBM → NVIDIA) does not restore HBM's tier. Because `output_share` is not consumed by scoring anywhere — cascade still walks `input_share`. That was the spec's deliberate scope on the two-field split ("Cascade and outbound criticality stay on `input_share` for now").

The paper's chokepoint framing of HBM ("sold out through 2026", "co-equal bottleneck with the GPU") is a volume + lead-time story, plus a customer-concentration story on the supplier side. Neither maps to inbound HHI on the HBM node. The two-field split correctly moved the customer-concentration piece into `output_share`, but the current scoring path does not read that field. When it does (its own task), HBM's tier can be re-examined.

Not adjusting.

8. Paper chokepoint tiers

node	normalized (default)	unnormalized
TSMC	critical (0.469)	critical (0.469)
ASML	high (0.202)	high (0.202)
gallium	critical (0.480)	critical (0.480)
dysprosium	critical (0.545)	critical (0.556)
HBM	high (0.178)	high (0.178)
CoWoS	critical (0.313)	critical (0.313)
RF & Power Semiconductors	critical (0.319)	critical (0.258)

Five of seven paper chokepoints in both modes. ASML and HBM read as high — both explained above. RF & Power holds critical on the paper-cited gallium 0.90 (§1A), not on tier-preservation

tuning.

9. Share-completeness backlog — grew, correctly

The backlog is larger than it was after the previous pass. Every re-authoring that lowered a `supplies/input_to` value pushed its bucket into the backlog. That is the deliverable, per spec (“Updated share-completeness backlog — expect it to grow, and that is correct”).

10. Test suite

normalize=true: 2 failed, 23 passed - `paper_chokepoints` — HBM and ASML at high, not critical. Explained in §6 and §7. - `share_completeness` — the backlog. Explained in §9.

normalize=false: 3 failed, 22 passed - Same two above. - `bounds` — `combined_hhi` (legacy blended value, inspection-only, not read by scoring) can exceed 1.0 for a handful of nodes. Known finding from the previous report; a two-line clamp available but out of scope.

11. Files changed

```
data/ai/edges.json           - 53 input_share re-authorings + 1
output_share
backend/tests/fixtures/ai/edges.json - fixture synced
```

Untouched: schema, scoring, config, narration, frontend. Thresholds unchanged, `normalize` default remains `true`. Cascade and outbound criticality continue reading `input_share`.